



CSE 623 Machine Learning

Weekly Report - 7

Group- Vector Minds

Team

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Project Title: Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

Project Definition: Agriculture plays an important role in our lives, especially in a well-known agricultural country like India. It provides food security to a larger population; however, its productivity depends heavily on factors such as rainfall and seasonal weather conditions. Due to increasing climate variability, predicting crop yields often becomes difficult.

The aim of this project is to develop a predictive system that estimates crop yield based on provided agricultural and historical data. The project applies a classical machine learning regression model to analyze the effect of environmental factors on agricultural productivity.

The project treats crop yield prediction as a supervised regression problem.

Where:

- Input variables: State, Crop type, Season, Crop Year, Area, Annual Rainfall, Fertilizer, Pesticide, and four engineered features — Fertilizer per Area, Pesticide per Area, Rainfall per Area, and Input Intensity
- Output variable (Target): Crop yield, which is production per unit area.

Objectives for Week 7

- Choose the best model from Random Forest and Gradient Boosting based on their tuning results, and prepare it for deployment.
- Undertake a feature importance analysis for the final model to determine the important features behind crop yield predictions.
- Determine the distribution of prediction errors from the final model to evaluate whether it is symmetrical and reliable in making predictions.
- Export the final model for later deployment using joblib, without retraining.
- Develop a prediction function that takes an input sample and uses the exported model to predict the crop yield.

Work Completed

This work involved choosing the best model from Week 6 tuning results, constructing the model explanation layer via feature importances evaluation, checking prediction errors, saving the model for deployment, and implementing a full-fledged real-world prediction tool.

Final Model Selection

The final model was chosen programmatically by comparing the `best_score_` attributes obtained using GridSearchCV when tuning Random Forest and Gradient Boosting pipelines in Week 6. Since Random Forest demonstrated higher cross-validated R^2 score, it was chosen as the final model. The model name was displayed to make sure about the right model choice and proceed to further steps. By doing so, we have excluded the assumptions about the model performance and objectively chosen the most accurate one based on its metrics.

Feature Importances

Feature importances of the final model were calculated using the `feature_importances_` attribute, accessed via the final tuned Random Forest pipeline steps. The feature importance bar plot included all input features, including those engineered in Week 5: Fertilizer per Area, Pesticide per Area, Rainfall per Area, and Input Intensity. The obtained plot confirmed once again that the important variables for prediction are: Area, Annual Rainfall, Fertilizer, and Crop Type. It is worth noting that the latter is not even the best predictor since there is no categorical encoding.

Distribution of Prediction Errors

Prediction errors were obtained by subtracting predicted from actual yield values in the test set. A histogram with a KDE plot overlay was created to visualize this distribution and then saved to the `graphs_week7` folder. The generated distribution was then checked for symmetry around zero. If the distribution is bell-shaped and centered around zero, then the trained model makes unbiased predictions; that is, it neither overpredicts nor underpredicts crop yield across different test cases.

Saving of Model

To make future predictions on unseen raw input data, the entire pipeline of the final model (i.e., the entire preprocessing steps along with the trained Random Forest model) was saved to disk in the file `final_crop_yield_model.pkl` using `joblib.dump()`. By saving the pipeline in this manner, one avoids the hassle of having to do the same preprocessing on new data in the future when making predictions, as well as avoid retraining the model.

Prediction Function

A reusable `predict_yield()` function was implemented which takes a dictionary of input feature values corresponding to a single agriculture record, converts it to a `DataFrame`, and then uses the pipeline stored on disk to make predictions about a crop yield value. Such a prediction function encapsulates the whole prediction process into a neat and convenient one-function call which can be seamlessly plugged into advisory apps or decision support systems used by farmers or government authorities.

Example Prediction

The developed prediction function was used with a sample input dataset for a crop of Rice harvested in Gujarat during the Kharif season of 2020, containing values for Area, Annual Rainfall, Fertilizer, Pesticide, and all four generated feature variables.

Final Observations

The seventh week marked the completion of transforming the experiment into a full-fledged crop yield predictor that can be deployed immediately. The final model was chosen objectively by evaluating its performance via cross-validation in the sixth week. The selected model was then evaluated using interpretability analysis by analyzing feature importance and verifying it via the error distribution test. Finally, the model was stored for future deployment. The predict function offers a clear and concise API endpoint, which makes it easy to reuse the code. Throughout the seven-week project, the whole process of machine learning was completed, including cleaning the data, performing exploratory analysis, and feature engineering in Weeks 5 and 6, tuning hyperparameters, cross-validation, and learning curves in Week 6, and finally, selecting the final model, interpretability, storing models, and deploying the application in Week 7.